

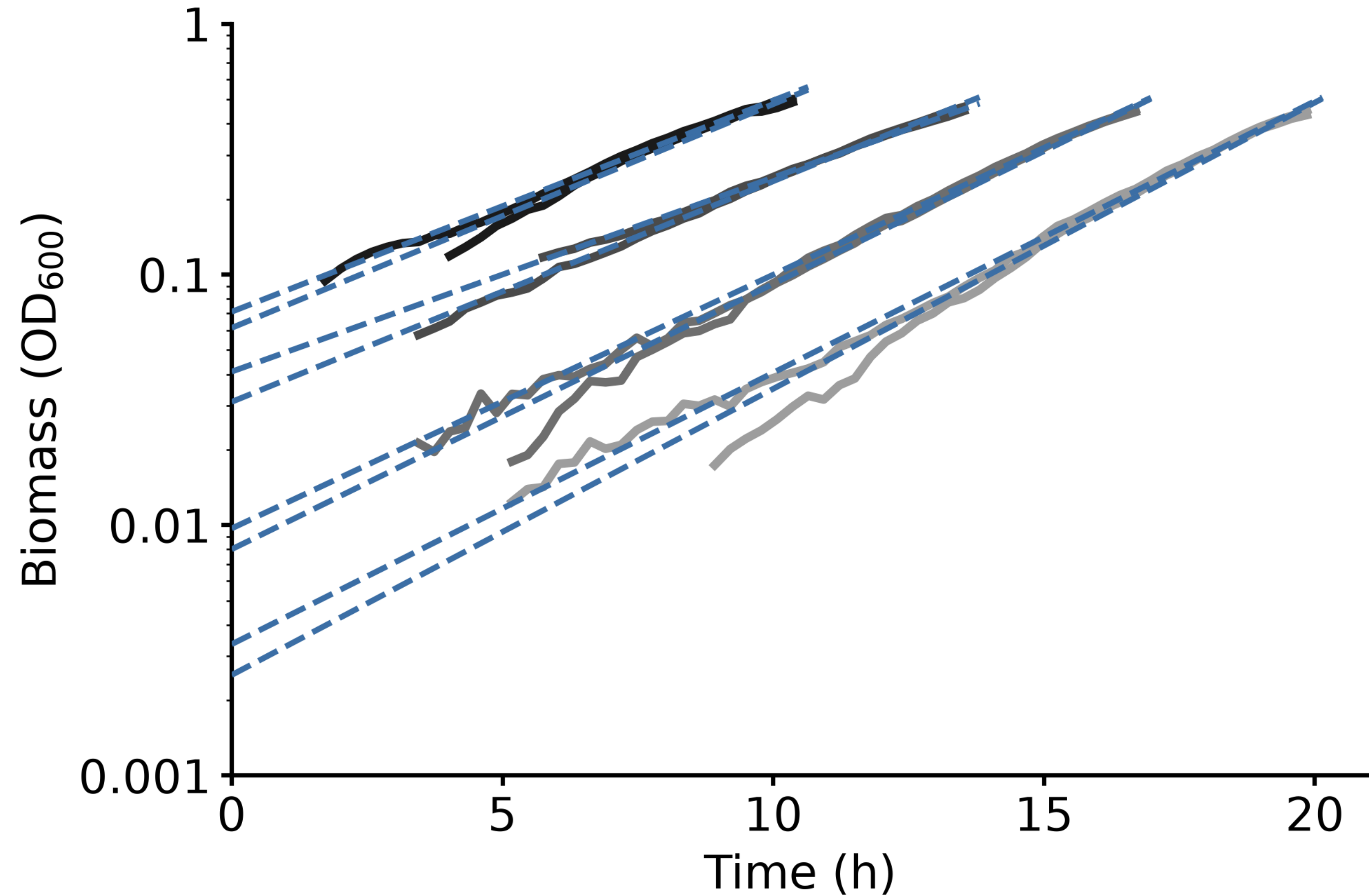
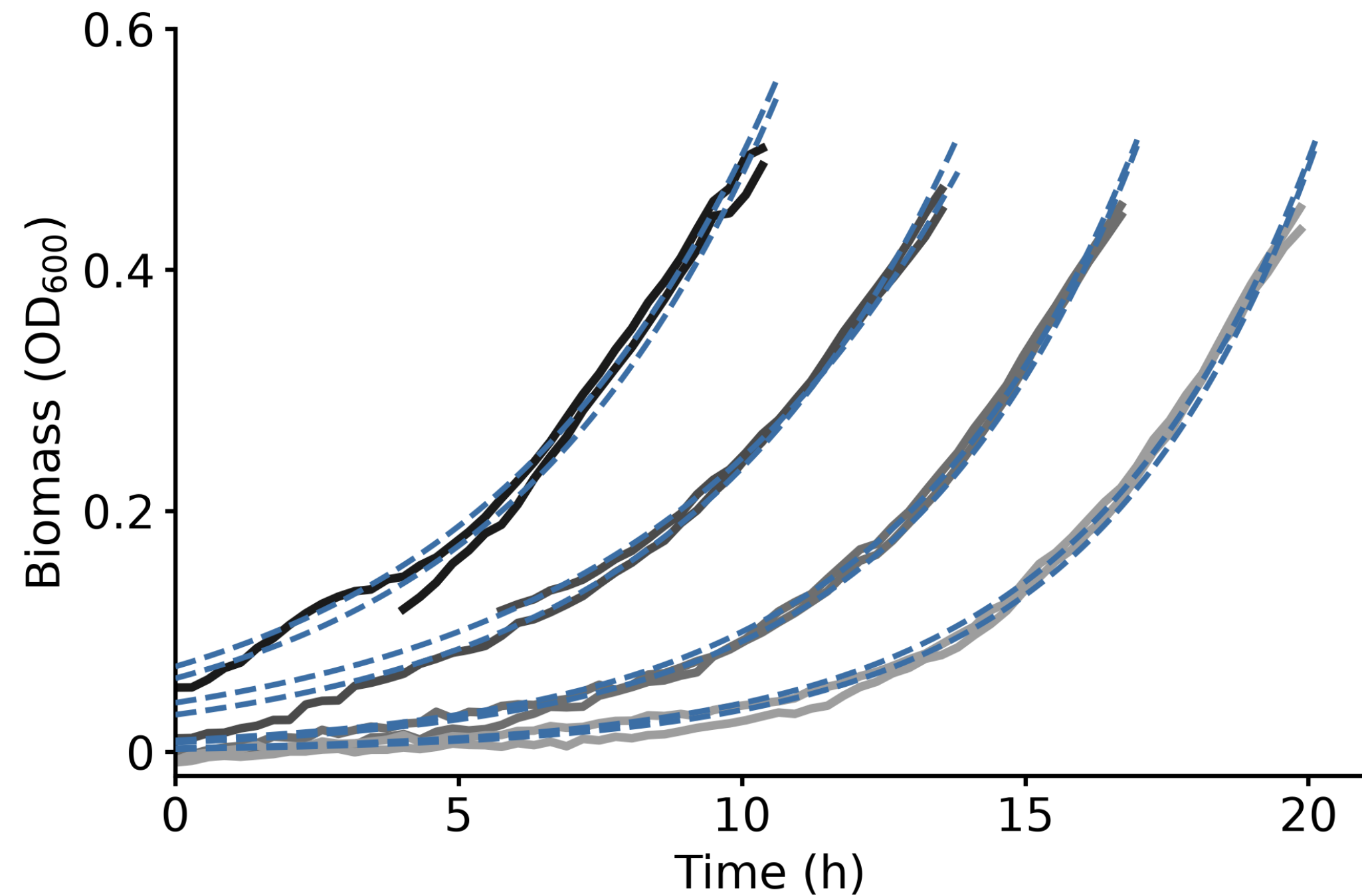
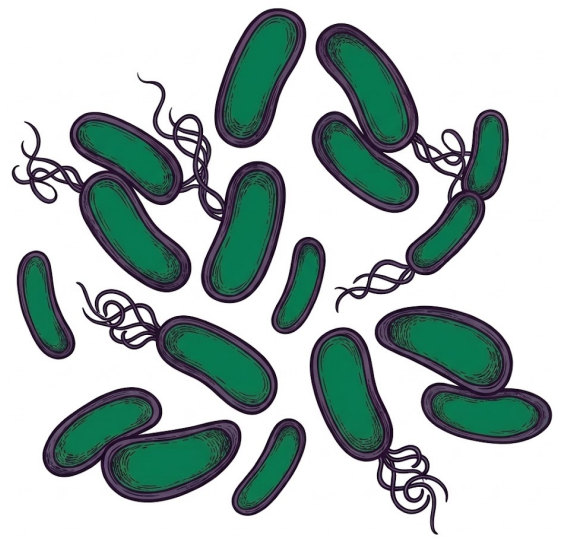
L1

Deterministic exponential growth

How does a population grow?



Are bacterial populations growing exponentially?

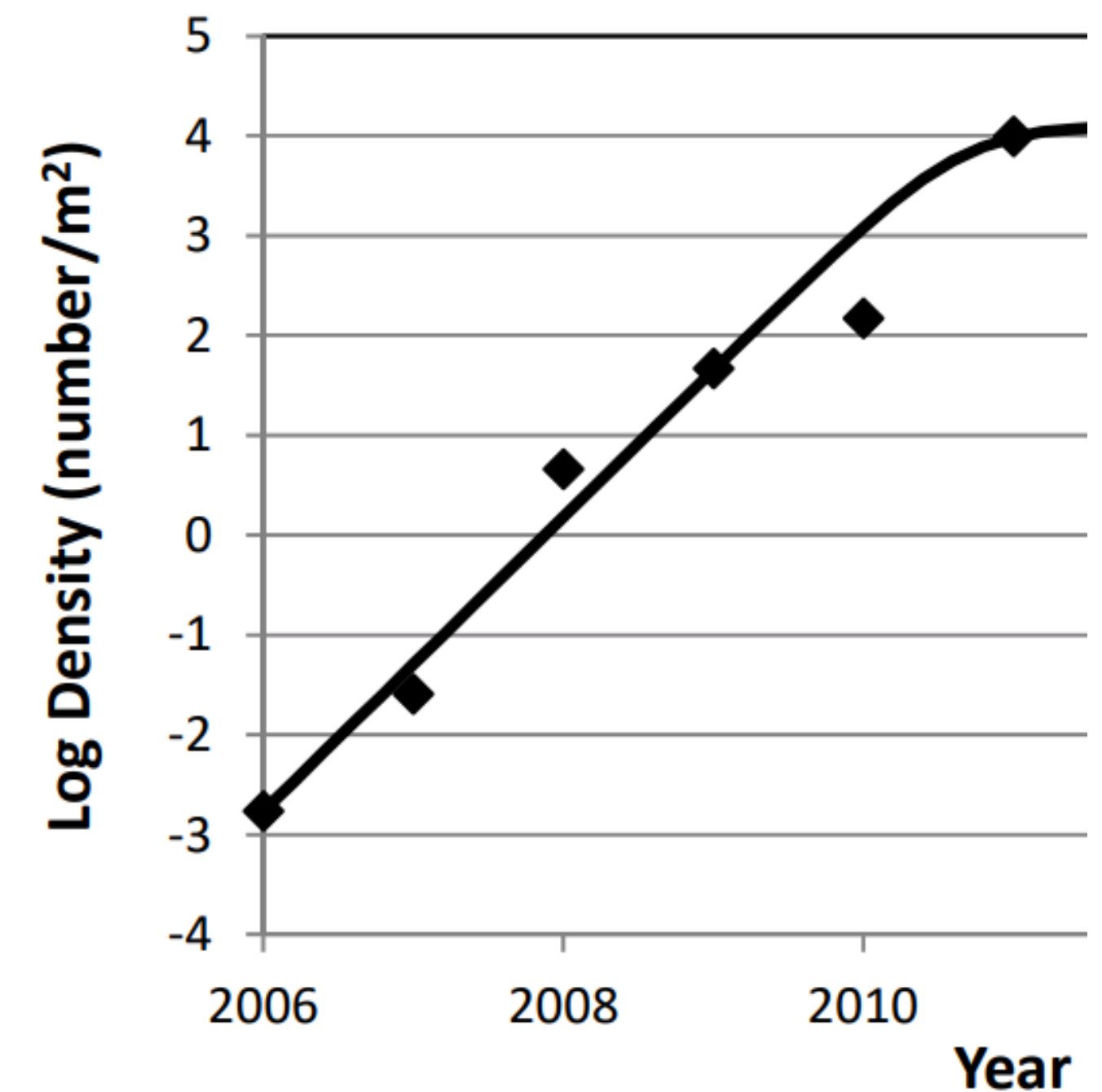
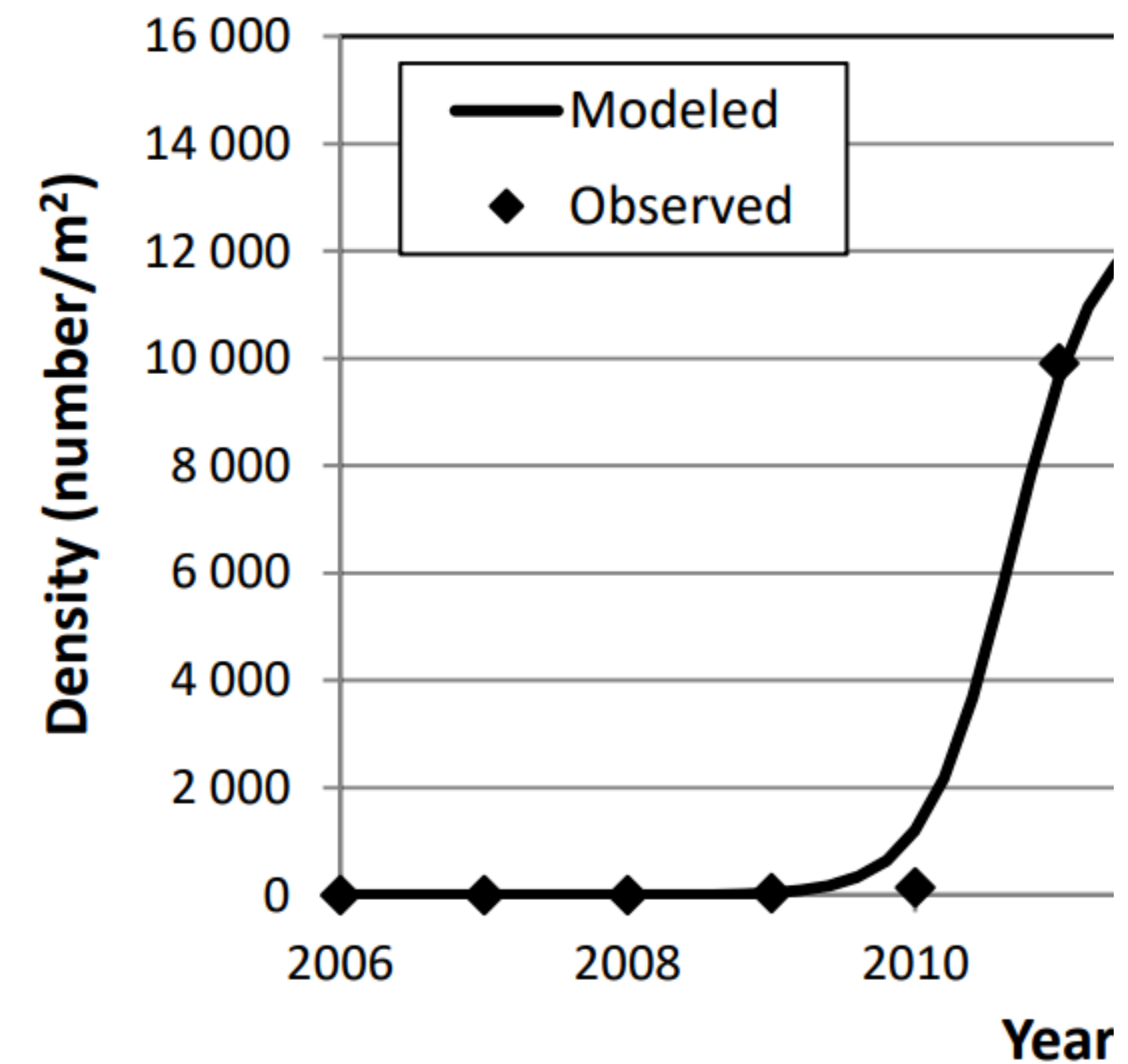


E. coli: size 10^{-6}m , generation time 10^4s (tens of mins)

Only bacteria?

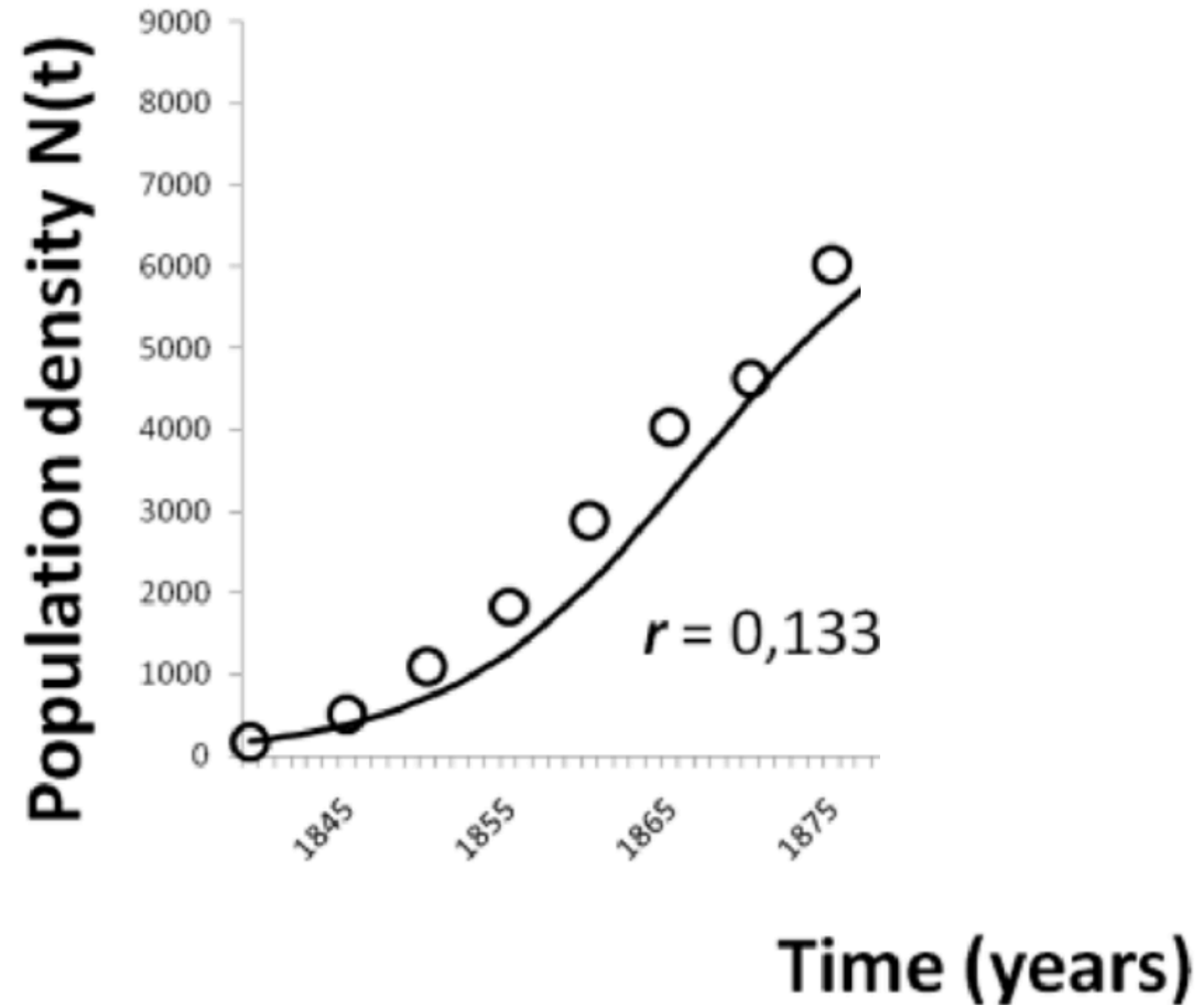


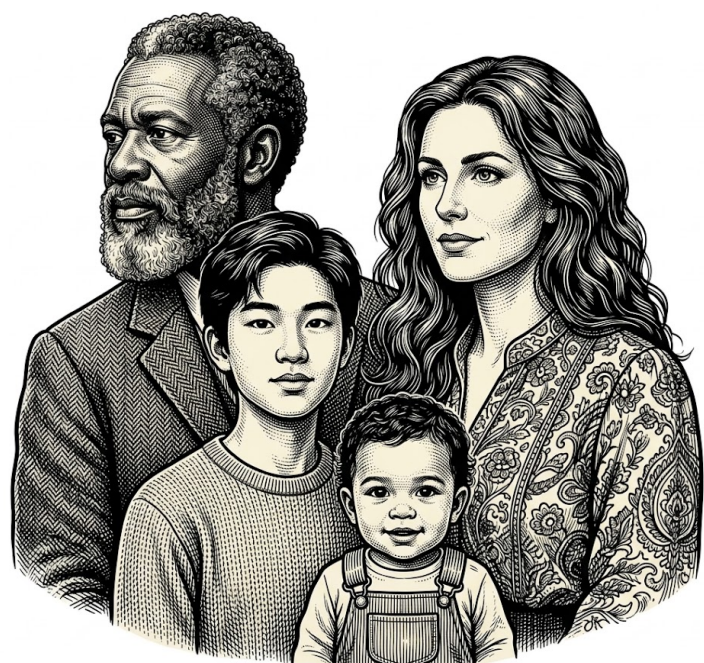
Zebra mussel:
size 10^{-1} m,
generation time 10^7 s (weeks)



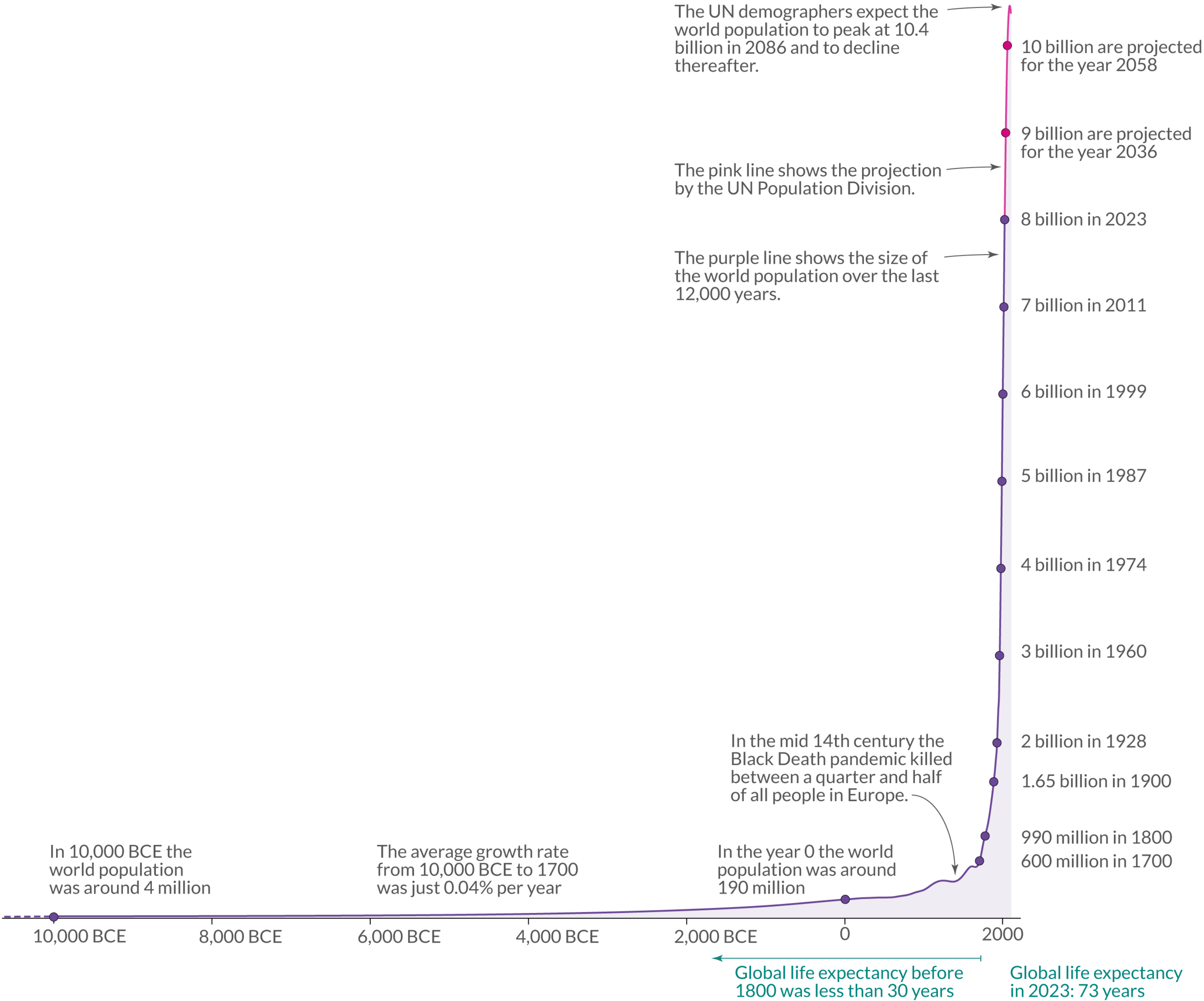
Only bacteria?

Sheep: size 10^0m , generation time 10^8s (year)



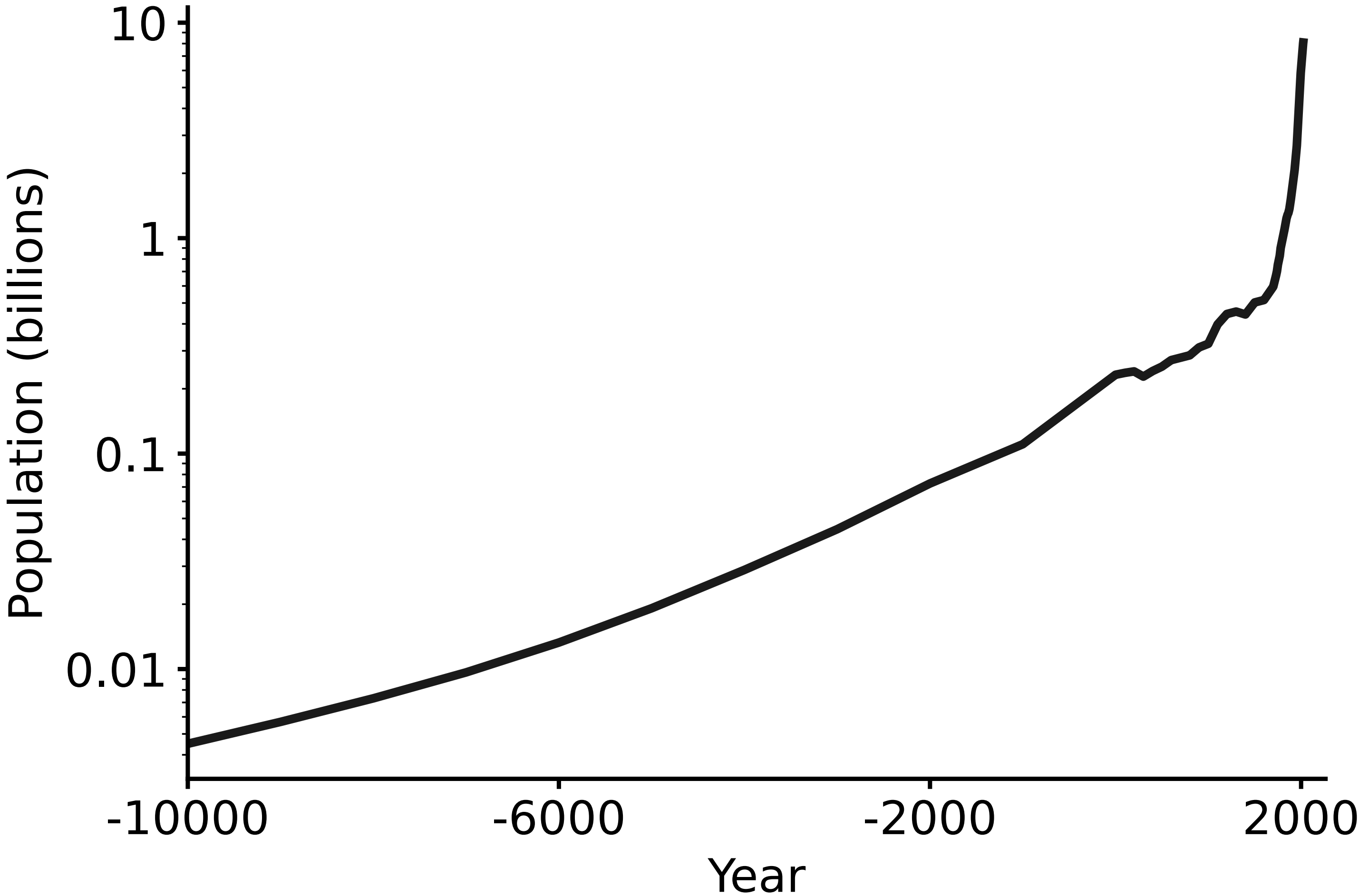


The size of the world population over the long-run

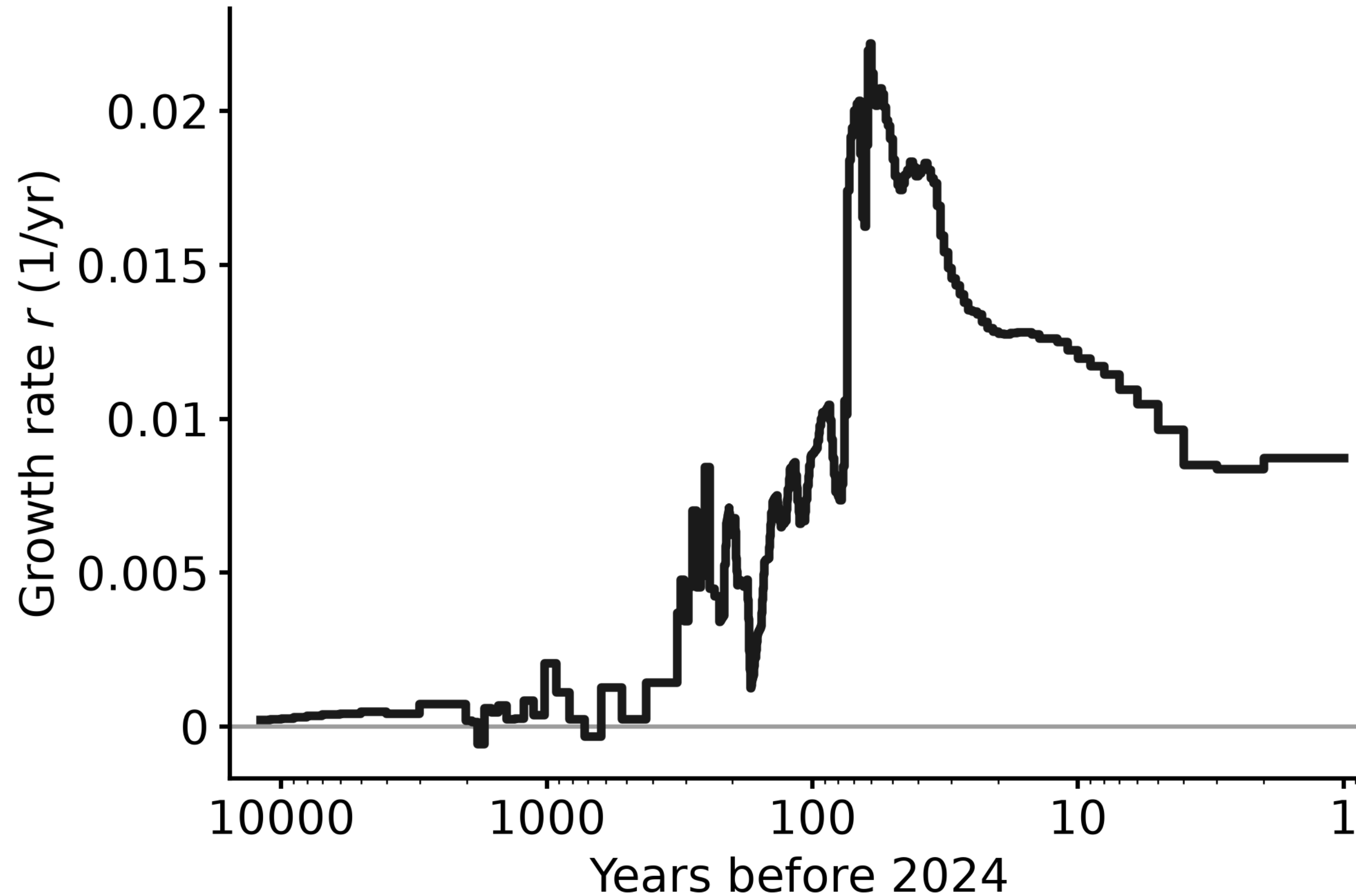


what fraction of ever-lived humans alive today?

Human population since 10000BC



Growth rate not really constant



what fraction of ever-lived humans alive today?

<https://ourworldindata.org/grapher/population>
download -> data

do it yourself